

INDU 211 · Production & Manufacturing Systems

Master Step-by-Step Quantitative Operations & Method Expansions Manual

Concordia University · Dept. of Mechanical, Industrial & Aerospace Engineering · Universal Expansion Standard

INDU 211: Introduction to Production and Manufacturing Systems

Master Step-by-Step Quantitative Operations & Method Expansions Manual

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Pedagogical Format Standard: Strict Step-by-Step Expansion with Zero Skipped Steps

Pedagogical Method & Formula Index

This manual applies the formal pedagogical step-by-step structure established in our master mathematics series to the quantitative engineering models of **INDU 211**. Every problem archetype follows four strict rules:

1. **Explicit Identification:** All given engineering parameters, economic coefficients, and constraints are declared with their exact physical units.
2. **Theory Before Algebra:** The governing governing law, economic definition, or geometric relationship is written in plain English before substituting numbers.
3. **Zero Skipped Calculations:** Every arithmetic step, intermediate fraction, conversion factor, and summation is written out completely.
4. **Pattern to Remember:** Every archetype concludes with an algorithmic memory box summarizing the decision rules for midterm and final exams.

Archetype 1: Single-Process Break-Even Analysis & Viability

Problem Statement

A precision aerospace manufacturing subcontractor is evaluating whether to acquire a computer-numerical-control (CNC) milling machine for a proprietary landing-gear titanium bracket. The accounting and industrial engineering departments provide the following annualized cost data:

- **Annual Fixed Overhead Cost (FC):** \$28,000\$ (covers equipment lease, dedicated clean-room space, and calibration contracts).
- **Variable Production Cost per Bracket (v):** \$100.00 / \text{bracket}\$ (direct aerospace alloy stock, CNC machinist wages, coolant, and cutting tool wear).
- **Contract Selling Price per Bracket (P):** \$200.00 / \text{bracket}\$.

Required Questions:

1. Calculate the annual Break-Even production volume (Q_{BEP}) in units.
2. Calculate the financial outcome (annual profit or net operating loss) if the market demand is only $Q = 220$ units/year. State whether management should proceed with equipment installation.

3. Calculate the annual operating profit if customer demand reaches $Q = 500$ units/year.
4. If market analysis reveals that demand cannot exceed $Q = 200$ units/year, calculate the minimum contract selling price ($P_{\min}$) required to avoid financial loss.

Step-by-Step Solution

Step 1: Identify all economic parameters and state the theoretical definitions of Total Cost, Total Revenue, and Operating Profit

- Fixed Capital Cost: $FC = 28,000\$$
- Unit Variable Cost: $v = 100.00 \text{ \$/unit}$
- Unit Selling Price: $P = 200.00 \text{ \$/unit}$
- Unit Contribution Margin: $CM = P - v$ (the gross margin each unit contributes toward covering fixed capital costs).

We define the primary cost and revenue functions as linear functions of production volume Q :

$$\text{Total Cost Function: } TC(Q) = FC + (v \cdot Q)$$

$$\text{Total Revenue Function: } TR(Q) = P \cdot Q$$

$$\text{Operating Profit Function: } \Pi(Q) = TR(Q) - TC(Q) = (P \cdot Q) - [FC + (v \cdot Q)] = (P - v)Q - FC$$

Step 2: Formulate and solve the Break-Even Condition

Theoretical Rule: The Break-Even Point (Q_{BEP}) is defined as the unique production volume where Total Revenue exactly equals Total Cost, resulting in zero net operating profit ($\Pi = 0$).

Set $TR(Q) = TC(Q)$:

$$P \cdot Q = FC + (v \cdot Q)$$

Subtract the total variable cost ($v \cdot Q$) from both sides of the equation:

$$P \cdot Q - v \cdot Q = FC$$

Factor out the common term Q :

$$Q \cdot (P - v) = FC$$

Divide both sides by the unit contribution margin ($P - v$):

$$Q_{BEP} = \frac{FC}{P - v}$$

Substitute the numerical parameters into the formula:

$$Q_{BEP} = \frac{28,000}{200 - 100}$$

Compute the denominator: $200 - 100 = 100 \text{ \$/unit}$

Perform the division:

$$Q_{BEP} = \frac{28,000}{100} = 280 \text{ units/year}$$

Conclusion: The facility must produce and sell exactly 280 brackets/year to completely recover all fixed overhead expenses.

Step 3: Compute operating profit/loss under constrained volume ($Q = 220$ units/year)

Theoretical Rule: Substitute $Q = 220$ directly into the operating profit equation $\Pi(Q) = (P - v)Q - FC$.

Calculate Total Revenue at $Q = 220$: $\$TR(220) = P \cdot Q = 200 \times 220 = \$44,000$

Calculate Total Variable Cost at $Q = 220$: $\$VC(220) = v \cdot Q = 100 \times 220 = \$22,000$

Calculate Total Cost at $Q = 220$: $\$TC(220) = FC + VC(220) = 28,000 + 22,000 = \$50,000$

Calculate Net Operating Profit $\Pi(220)$: $\Pi(220) = TR(220) - TC(220) = 44,000 - 50,000 = -\$6,000$

Alternatively, using the contribution margin formula: $\Pi(220) = (200 - 100)(220) - 28,000 = (100)(220) - 28,000 = 22,000 - 28,000 = -\$6,000$

Managerial Decision: Operating at $Q = 220$ units results in an **annual net loss of $-\$6,000$** . Management must **reject installation** under this demand projection because the volume is strictly below the break-even threshold ($220 < 280$).

Step 4: Compute operating profit at expanded volume ($Q = 500$ units/year)

Substitute $Q = 500$ into the profit formula:

$$\Pi(500) = (P - v)Q - FC$$

$$\Pi(500) = (200 - 100)(500) - 28,000$$

$$\Pi(500) = (100 \times 500) - 28,000$$

$$\Pi(500) = 50,000 - 28,000 = +\$22,000$$

Managerial Decision: At 500 units/year, the process generates a robust **annual operating profit of $+\$22,000$** .

Step 5: Reverse engineer the minimum selling price ($P_{\min}$) for $Q = 200$ units

Theoretical Rule: Set the operating profit $\Pi(Q) = 0$ at the fixed target volume $Q = 200$, treating price P as the algebraic unknown:

$$(P - v)Q - FC = 0$$

Add FC to both sides:

$$(P - v)Q = FC$$

Divide both sides by Q :

$$P - v = \frac{FC}{Q}$$

Add v to both sides to isolate P :

$$P_{\min} = v + \frac{FC}{Q}$$

Substitute $FC = 28,000$, $v = 100$, and $Q = 200$:

$$P_{\min} = 100 + \frac{28,000}{200}$$

Compute the fraction:

$$\frac{28,000}{200} = \frac{280}{2} = 140$$

Perform the addition: $P_{\min} = 100 + 140 = \$240.00$

Conclusion: If market volume cannot exceed 200 units, management must negotiate a contract price of at least **$\$240.00$** to break even.

Pattern to Remember: Single-Process Break-Even

1. Contribution margin represents the cash per unit available to pay down fixed costs: $CM = P - v$.
2. Break-Even Volume formula: $Q_{BEP} = \frac{FC}{P-v} = \frac{FC}{CM}$.
3. If $Q < Q_{BEP} \implies$ Operating Loss. If $Q > Q_{BEP} \implies$ Operating Profit.
4. To break even at a lower volume, you must either decrease fixed costs, decrease variable unit costs, or raise the selling price: $P_{\min} = v + \frac{FC}{Q}$.

Archetype 2: Multi-Process Selection & Crossover Volume Analysis

Problem Statement

A manufacturing plant needs to produce a standardized stamped automotive bracket. The plant manager is evaluating three distinct manufacturing process alternatives:

- **Process A (Manual Job-Shop):**
 - Annual Fixed Cost: $FC_A = 10,000\$$
 - Variable Cost per unit: $v_A = 15.00/\text{unit}\$$
- **Process B (Semi-Automated Cellular Manufacturing):**
 - Annual Fixed Cost: $FC_B = 40,000\$$
 - Variable Cost per unit: $v_B = 5.00/\text{unit}\$$
- **Process C (Fully Automated Transfer Line / Robotics):**
 - Annual Fixed Cost: $FC_C = 100,000\$$
 - Variable Cost per unit: $v_C = 2.00/\text{unit}\$$

Required Questions:

1. State the Total Cost function for each of the three candidate processes.
2. Determine the algebraic Crossover Volume ($Q_{A,B}$) between Process A and Process B.
3. Determine the algebraic Crossover Volume ($Q_{B,C}$) between Process B and Process C.

4. Establish the comprehensive piecewise decision rule specifying which process minimizes total manufacturing cost across all possible annual volumes $Q \geq 0$.
5. If the forecasted market volume is $Q = 12,000$ units/year, verify algebraically which process delivers the lowest cost.

Step-by-Step Solution

Step 1: Formulate the Total Cost functions

Theoretical Rule: The Total Cost $TC_i(Q)$ for any process i is the linear sum of its annualized fixed investment and volume-dependent variable operating expenses:

$$TC_i(Q) = FC_i + v_i \cdot Q$$

Writing out the explicit equation for each alternative:

$$TC_A(Q) = 10,000 + 15Q$$

$$TC_B(Q) = 40,000 + 5Q$$

$$TC_C(Q) = 100,000 + 2Q$$

Step 2: Calculate the Crossover Volume between Process A and Process B ($Q_{A,B}$)

Theoretical Rule: The crossover point between two processes is the production volume where both processes incur identical total operating costs:

$$TC_A(Q) = TC_B(Q)$$

Equate the two functions:

$$10,000 + 15Q = 40,000 + 5Q$$

Collect variable cost terms on the left side by subtracting $5Q$ from both sides:

$$10,000 + 15Q - 5Q = 40,000$$

$$10,000 + 10Q = 40,000$$

Collect fixed cost terms on the right side by subtracting 10,000 from both sides:

$$10Q = 40,000 - 10,000$$

$$10Q = 30,000$$

Divide by the difference in variable costs:

$$Q_{A,B} = \frac{30,000}{10} = 3,000 \text{ units/year}$$

Verification of equality at $Q = 3,000$: $\$TC_A(3,000) = 10,000 + (15 \times 3,000) = 10,000 + 45,000 = \$55,000$
 $\$TC_B(3,000) = 40,000 + (5 \times 3,000) = 40,000 + 15,000 = \$55,000$ Equality holds perfectly ($\$55,000 = \$55,000$).

Step 3: Calculate the Crossover Volume between Process B and Process C ($Q_{B,C}$)

Equate the total costs of Process B and Process C:

$$TC_B(Q) = TC_C(Q)$$

$$40,000 + 5Q = 100,000 + 2Q$$

Subtract $2Q$ from both sides:

$$40,000 + 3Q = 100,000$$

Subtract 40,000 from both sides:

$$3Q = 100,000 - 40,000$$

$$3Q = 60,000$$

Divide by 3:

$$Q_{B,C} = \frac{60,000}{3} = 20,000 \text{ units/year}$$

Verification of equality at $Q = 20,000$: $TC_B(20,000) = 40,000 + (5 \times 20,000) = 40,000 + 100,000 = \$140,000$
 $TC_C(20,000) = 100,000 + (2 \times 20,000) = 100,000 + 40,000 = \$140,000$ Equality holds perfectly ($\$140,000 = \$140,000$).

Step 4: Check for Process Dominance and formulate the Piecewise Decision Rule

Notice that the crossover quantities are strictly ascending:

$$Q_{A,B} = 3,000 < Q_{B,C} = 20,000$$

This confirms that Process B is not dominated and possesses an active economic range where it is the strictly cheapest option.

Formal Piecewise Selection Rule:

- **Interval 1: For $0 \leq Q < 3,000$ units/year:**
 - Select **Process A (Manual Job-Shop)**. At low production volumes, low fixed capital cost dominates, and high variable cost is minimized.
 - **Interval 2: For $3,000 < Q < 20,000$ units/year:**
 - Select **Process B (Semi-Automated Cellular)**. The higher fixed capital of $\$40,000$ is effectively amortized by the lower $\$5/\text{unit}$ variable cost.
 - **Interval 3: For $Q > 20,000$ units/year:**
 - Select **Process C (Fully Automated Robotics)**. Mass-production volume fully absorbs the massive $\$100,000$ fixed tooling cost, leveraging the rock-bottom $\$2.00/\text{unit}$ variable cost.
 - **At Transition Boundaries ($Q = 3,000$ and $Q = 20,000$):**
 - Management is economically indifferent between the two adjacent alternatives based on cost alone (non-cost factors such as flexibility or ramp-up speed decide).
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Step 5: Quantitative Evaluation at Projected Volume $Q = 12,000$ units/year

Theoretical Rule: Compare the total costs of all three options at $Q = 12,000$:

- **Process A:** $TC_A(12,000) = 10,000 + 15(12,000) = 10,000 + 180,000 = \$190,000$

- **Process B:** $TC_B(12,000) = 40,000 + 5(12,000) = 40,000 + 60,000 = \$100,000$
- **Process C:** $TC_C(12,000) = 100,000 + 2(12,000) = 100,000 + 24,000 = \$124,000$

Comparing the three total cost figures: $\$100,000 \text{ (B)} < \$124,000 \text{ (C)} < \$190,000 \text{ (A)}$

Conclusion: At $Q = 12,000$ units/year, **Process B is the optimal choice**, delivering a net annual cost savings of **\$24,000** over Process C and **\$90,000** over Process A.

Pattern to Remember: Multi-Process Crossover

1. General crossover formula between any two processes 1 and 2:

$$Q_{1,2} = \frac{FC_2 - FC_1}{v_1 - v_2}$$

2. Order the processes by increasing fixed cost: $FC_A < FC_B < FC_C$.
3. Verify that the variable costs decrease in reverse order: $v_A > v_B > v_C$.
4. If $Q_{A,B} < Q_{B,C}$, all three processes have an active economic operating window. If $Q_{A,B} > Q_{B,C}$, the intermediate process (B) is economically dominated and should never be chosen!

Archetype 3: Multi-Facility Center of Gravity (CoG) & Spatial Distance Metrics

Problem Statement

A regional supply chain director must establish the optimal geographic location (x^*, y^*) for a central distribution center (CDC) to supply four retail superstores (S_1, S_2, S_3, S_4) situated across an industrial corridor mapped onto a Cartesian coordinate grid (measured in kilometers):

- **Store S_1 :** Located at $(x_1, y_1) = (2, 2)$, with weekly shipment demand $W_1 = 800$ pallets/week.
- **Store S_2 :** Located at $(x_2, y_2) = (3, 5)$, with weekly shipment demand $W_2 = 900$ pallets/week.
- **Store S_3 :** Located at $(x_3, y_3) = (5, 4)$, with weekly shipment demand $W_3 = 200$ pallets/week.
- **Store S_4 :** Located at $(x_4, y_4) = (8, 5)$, with weekly shipment demand $W_4 = 100$ pallets/week.

Required Questions:

1. Explain the physical and engineering distinction between Euclidean distance (L_2) and Rectilinear distance (L_1), stating when each metric is applied.
2. Calculate the **Unweighted Centroid** $(\bar{x}_{unw}, \bar{y}_{unw})$ of the four stores.
3. Calculate the **Volume-Weighted Center of Gravity** (x^*, y^*) .
4. Compute the total rectilinear travel distance from the optimal weighted CDC location to Store S_1 and Store S_4 .
5. Provide the engineering and economic rationale explaining why (x^*, y^*) *diverges substantially from* $(\bar{x}_{unw}, \bar{y}_{unw})$.

Step-by-Step Solution

Step 1: Define Spatial Distance Metrics (Euclidean vs Rectilinear)

Theoretical Definition:

- **Euclidean Distance (L_2 Metric - Straight-Line Distance):**

$$d_E(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- *Application:* Used for direct airborne freight (drones, cargo planes), cross-country oil pipelines, line-of-sight telecommunications microwave links, or high-sea shipping routes where travel is unimpeded by ground obstacles.

- **Rectilinear Distance (L_1 Metric - Manhattan / City-Block / Orthogonal Grid):**

$$d_R(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$$

- *Application:* Used for urban truck delivery fleets navigating perpendicular street grids, factory floor automated guided vehicles (AGVs) navigating rectangular aisle networks, and microchip wire routing.

Step 2: Compute the Unweighted Geometric Centroid $(\bar{x}_{unw}, \bar{y}_{unw})$

Theoretical Rule: The unweighted centroid represents the purely geometric mean of the coordinates, treating every customer node with equal importance ($n = 4$): $\bar{x}_{unw} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + x_3 + x_4}{4}$

$$\bar{y}_{unw} = \frac{\sum_{i=1}^n y_i}{n} = \frac{y_1 + y_2 + y_3 + y_4}{4}$$

Compute the x-coordinate:

$$\bar{x}_{unw} = \frac{2 + 3 + 5 + 8}{4} = \frac{18}{4} = 4.50 \text{ km}$$

Compute the y-coordinate:

$$\bar{y}_{unw} = \frac{2 + 5 + 4 + 5}{4} = \frac{16}{4} = 4.00 \text{ km}$$

Result: The unweighted geometric center is located at coordinates (4.50, 4.00).

Step 3: Compute the Volume-Weighted Center of Gravity (x^*, y^*)

Theoretical Rule: The weighted center of gravity minimizes the total sum of squared Euclidean transport-work ($W_i \cdot d_i^2$).

The optimal coordinates (x^*, y^*) are given by the weighted averages: $x^* = \frac{\sum_{i=1}^n W_i x_i}{\sum_{i=1}^n W_i}$, $y^* = \frac{\sum_{i=1}^n W_i y_i}{\sum_{i=1}^n W_i}$

Part A: Compute the Total Demand Weight ($\sum W_i$)

$$\sum_{i=1}^4 W_i = W_1 + W_2 + W_3 + W_4$$

$$\sum_{i=1}^4 W_i = 800 + 900 + 200 + 100 = 2,000 \text{ pallets/week}$$

Part B: Formulate and Compute the Weighted Sum for X ($\sum W_i x_i$)

$$\sum_{i=1}^4 W_i x_i = (W_1 \cdot x_1) + (W_2 \cdot x_2) + (W_3 \cdot x_3) + (W_4 \cdot x_4)$$

$$\sum_{i=1}^4 W_i x_i = (800 \times 2) + (900 \times 3) + (200 \times 5) + (100 \times 8)$$

$$\sum_{i=1}^4 W_i x_i = 1,600 + 2,700 + 1,000 + 800 = 6,100 \text{ pallet} \cdot \text{km}$$

$$\text{Calculate } x^* = \frac{6,100}{2,000} = \frac{61}{20} = 3.05 \text{ km}$$

Part C: Formulate and Compute the Weighted Sum for Y ($\sum W_i y_i$)

$$\sum_{i=1}^4 W_i y_i = (W_1 \cdot y_1) + (W_2 \cdot y_2) + (W_3 \cdot y_3) + (W_4 \cdot y_4)$$

$$\sum_{i=1}^4 W_i y_i = (800 \times 2) + (900 \times 5) + (200 \times 4) + (100 \times 5)$$

$$\sum_{i=1}^4 W_i y_i = 1,600 + 4,500 + 800 + 500 = 7,400 \text{ pallet} \cdot \text{km}$$

$$\text{Calculate } y^* = \frac{7,400}{2,000} = \frac{74}{20} = 3.70 \text{ km}$$

Result: The optimal volume-weighted Center of Gravity coordinates are $(x^*, y^*) = (3.05, 3.70)$.

Step 4: Compute Rectilinear Travel Distance to S_1 and S_4

Using the Manhattan metric $d_R = |x^* - x_i| + |y^* - y_i|$ from the weighted warehouse at $(3.05, 3.70)$:

To Store $S_1(2, 2)$:

$$d_R(S_1) = |3.05 - 2| + |3.70 - 2| = 1.05 + 1.70 = 2.75 \text{ km}$$

To Store $S_4(8, 5)$:

$$d_R(S_4) = |3.05 - 8| + |3.70 - 5| = |-4.95| + |-1.30| = 4.95 + 1.30 = 6.25 \text{ km}$$

Step 5: Engineering & Economic Sensitivity Analysis

Comparing the two solutions:

- Unweighted centroid: $(4.50, 4.00)$
- Weighted center of gravity: $(3.05, 3.70)$

Why did the location shift? Notice that stores S_1 (800 pallets) and S_2 (900 pallets) together account for $\frac{800+900}{2,000} = \frac{1,700}{2,000} = 85$ of the total weekly delivery volume. Because transportation cost is proportional to **Weight** $\times$ **Distance**, the optimization algorithm shifts the warehouse strongly toward the high-volume cluster ($x \approx 2$ to 3), rather than placing it near the distant low-volume store S_4 at $x = 8$ (which accounts for only 5 of total shipments).

Pattern to Remember: Facility Location & Center of Gravity

1. Always sum total volume first: $W_{total} = \sum W_i$.
2. Multiply each node coordinate by its respective weight: $W_i x_i$ and $W_i y_i$.
3. Divide weighted sums by total volume:

$$x^* = \frac{\sum W_i x_i}{\sum W_i}, \quad y^* = \frac{\sum W_i y_i}{\sum W_i}$$

4. Distance check:

- Euclidean (straight line): $d_E = \sqrt{\Delta x^2 + \Delta y^2}$
- Rectilinear (city grid): $d_R = |\Delta x| + |\Delta y|$

Archetype 4: Manufacturing Cycle Time, Production Rate & Operational Availability

Problem Statement

A dedicated automated manufacturing cell produces transmission shafts for heavy commercial vehicles. The cell operates on a standard **8-hour working shift** (480 minutes). The industrial engineering time-study records the following operational parameters per shaft:

- **Basic Machining Time (T_m):** 1.80 min/piece
- **Workpart Handling / Clamping Time (T_h):** 0.60 min/piece
- **Tool Replacement Time Allocation (T_t):** 0.10 min/piece (representing cutting insert indexation amortized per piece).

During a typical shift, the plant schedule includes:

- **Planned Downtime:** 30 minutes allocated for operator lunch and scheduled preventative maintenance.
- **Unplanned Downtime:** Due to chip clearing, sensor calibration faults, and tool jams, the machine experiences an average of 45 minutes of unexpected stoppage per shift.

Required Questions:

1. Determine the total operational Cycle Time (T_c) per piece in minutes and seconds.
2. Determine the cell's theoretical Hourly Production Rate (R_p) in pieces per hour.
3. Calculate the Machine Availability (A) for the shift.
4. Calculate the cell's actual net production output (Q_{actual}) in parts completed per 8-hour shift.

Step-by-Step Solution

Step 1: Identify and define all production cycle time components

Theoretical Rule: The operational cycle time (T_c) is the total time elapsed from the moment one part enters the workstation until the next consecutive part is ready to begin:

$$T_c = T_m + T_h + T_t$$

Where:

- T_m : Actual cutting or processing time (1.80 min).

- T_h : Material handling, loading, orienting, and unloading time (0.60 min).
- T_t : Tool service time allocated per part (0.10 min).

Substitute the numerical values:

$$T_c = 1.80 + 0.60 + 0.10 = 2.50 \text{ minutes/piece}$$

Convert decimal minutes to minutes and seconds:

$$0.50 \text{ min} = 0.50 \times 60 \text{ seconds} = 30 \text{ seconds}$$

$$T_c = \mathbf{2 \text{ minutes and } 30 \text{ seconds/piece}}$$

Step 2: Compute the theoretical Hourly Production Rate (R_p)

Theoretical Rule: The production rate R_p is the reciprocal of cycle time, scaled to one hour (60 minutes):

$$R_p = \frac{60 \text{ minutes/hour}}{T_c \text{ minutes/piece}}$$

Substitute $T_c = 2.50 \text{ min/piece}$:

$$R_p = \frac{60}{2.50} = \frac{60}{\frac{5}{2}} = 60 \times \frac{2}{5} = \frac{120}{5} = 24 \text{ pieces/hour}$$

Result: Under continuous 100% operation, the cell produces exactly 24 finished transmission shafts per hour.

Step 3: Compute Machine Availability (A)

Theoretical Rule: Machine Availability measures the fraction of planned operating time during which the equipment is physically running and operational:

$$A = \frac{\text{Actual Operating Time}}{\text{Planned Production Time}} = \frac{\text{Total Shift Time} - \text{Planned Downtime} - \text{Unplanned Downtime}}{\text{Total Shift Time} - \text{Planned Downtime}}$$

Part A: Compute Total Shift Time (T_{shift})

$$T_{shift} = 8 \text{ hours} \times 60 \text{ min/hour} = 480 \text{ minutes}$$

Part B: Compute Planned Production Time ($T_{planned}$)

Planned production time excludes scheduled lunch and preventative maintenance:

$$T_{planned} = T_{shift} - \text{Planned Downtime} = 480 - 30 = 450 \text{ minutes}$$

Part C: Compute Actual Net Operating Time ($T_{operating}$)

Operating time is the remaining time the spindle is actually rotating after unplanned equipment breakdowns:

$$T_{operating} = T_{planned} - \text{Unplanned Downtime} = 450 - 45 = 405 \text{ minutes}$$

Part D: Calculate Availability Ratio (A)

$$A = \frac{T_{operating}}{T_{planned}} = \frac{405}{450}$$

Divide numerator and denominator by 45: $A = \frac{405 / 45}{450 / 45} = \frac{9}{10} = 0.90 \text{ or } \mathbf{90.0\%}$

Result: The automated cell exhibits an availability of 90.0.

Step 4: Calculate Actual Net Production Output per Shift (Q_{actual})

Theoretical Rule: Actual output is the product of theoretical hourly production rate and total operational hours, or simply the operating time in minutes divided by cycle time:

$$Q_{\text{actual}} = R_p \times \left(\frac{T_{operating}}{60} \right) = \frac{T_{operating}}{T_c}$$

Method 1 (Using cycle time):

$$Q_{\text{actual}} = \frac{405 \text{ minutes}}{2.50 \text{ minutes/piece}} = \frac{405}{\frac{5}{2}} = \frac{810}{5} = 162 \text{ pieces/shift}$$

Method 2 (Using hourly rate and available hours):

$$\text{Operating Hours} = \frac{405}{60} = 6.75 \text{ hours}$$

$$Q_{\text{actual}} = 24 \text{ pieces/hour} \times 6.75 \text{ hours} = 162 \text{ pieces/shift}$$

Result: The cell successfully produces 162 finished shafts during the 8-hour shift.

Pattern to Remember: Cycle Time & Availability

1. Total Unit Cycle Time: $T_c = T_m + T_h + T_t$ (Machining + Handling + Tool service).
2. Hourly Production Rate: $R_p = \frac{60}{T_c}$ (pieces/hour).
3. Availability Formula:

$$A = \frac{T_{operating}}{T_{planned}} = \frac{\text{Planned Time} - \text{Breakdowns}}{\text{Planned Time}}$$

4. Actual Shift Output:

$$Q = \frac{T_{operating}}{T_c} = R_p \times \text{Hours Operated}$$

Archetype 5: Assembly Line Balancing & Minimum Workstation Formulation

Problem Statement

An industrial appliance manufacturer is configuring a manual progressive assembly line to produce electric lawn mowers. The target production quota is $Q = 480$ units/day operating on a single 8-hour shift (480 minutes/day). The product structure consists of 8 elemental assembly tasks with a combined total work content time:

$$T_{wc} = \sum_{k=1}^8 t_k = 4.00 \text{ minutes/unit}$$

The line designer initially proposes a configuration utilizing $N = 6$ sequential workstations.

Required Questions:

1. Calculate the required line Cycle Time (T_c) to achieve the daily production quota.
2. Determine the Theoretical Minimum Number of Workstations ($N_{\min}$).
3. Calculate the Assembly Line Balancing Efficiency (E) for the proposed $N = 6$ workstation layout.
4. Calculate the Total Balance Delay (d).

Step-by-Step Solution

Step 1: Calculate the required line Cycle Time (T_c)

Theoretical Rule: The line cycle time represents the maximum allowable time a unit can remain at any workstation to meet the daily target volume Q :

$$T_c = \frac{\text{Available Daily Operating Time}}{\text{Target Daily Output Volume}} = \frac{T_{avail}}{Q}$$

Substitute $T_{avail} = 480$ minutes/day and $Q = 480$ units/day:

$$T_c = \frac{480 \text{ minutes}}{480 \text{ units}} = 1.00 \text{ minute/unit} = 60 \text{ seconds/unit}$$

Result: Every workstation on the assembly line must complete its assigned tasks within 1.00 minute (60 seconds).

Step 2: Determine Theoretical Minimum Workstations ($N_{\min}$)

Theoretical Rule: The theoretical minimum number of stations $N_{\min}$ is the total work content time divided by the cycle time, rounded up to the nearest integer ($\lceil \cdot \rceil$):

$$N_{\min} = \left\lceil \frac{T_{wc}}{T_c} \right\rceil$$

Substitute $T_{wc} = 4.00$ minutes and $T_c = 1.00$ minute:

$$\frac{T_{wc}}{T_c} = \frac{4.00}{1.00} = 4.00$$

Since 4.00 is already an integer:

$$N_{\min} = 4 \text{ workstations}$$

Result: In an ideal, perfectly balanced assembly line with zero idle time, exactly **4 workstations** are required.

Step 3: Compute Line Balancing Efficiency (E) for N = 6 workstations

Theoretical Rule: Line Efficiency is the ratio of total productive work content to total available line capacity:

$$E = \frac{T_{wc}}{N \cdot T_c} \times 100$$

Where:

- $T_{wc} = 4.00$ minutes
- $N = 6$ workstations
- $T_c = 1.00$ minute/station

Calculate total available workstation time capacity:

$$N \cdot T_c = 6 \times 1.00 = 6.00 \text{ minutes/unit}$$

Calculate Efficiency E : $E = \frac{4.00}{6.00} = \frac{2}{3} \approx 0.6667$ or $\mathbf{66.67\%}$

Result: The proposed 6-station line operates at **66.67 line efficiency**.

Step 4: Compute Total Balance Delay (d)

Theoretical Rule: Balance Delay (also called Idle Time Percentage) is the fraction of workstation time lost to worker starvation or blocking:

$$d = 1 - E = 100$$

Alternatively:

$$d = \frac{N \cdot T_c - T_{wc}}{N \cdot T_c} = \frac{6.00 - 4.00}{6.00} = \frac{2.00}{6.00} = \frac{1}{3} \approx 33.33$$

Managerial Insight: One-third (33.33) of total paid labor time on this 6-station line is spent completely idle. An industrial engineer should rebalance tasks or combine operations closer to $N = 4$ or $N = 5$ to increase labor productivity.

Pattern to Remember: Assembly Line Balancing

1. Required Cycle Time: $T_c = \frac{\text{Available Operating Time}}{\text{Demand Volume } Q}$.
2. Minimum Possible Stations: $N_{\min} = \left\lceil \frac{T_{wc}}{T_c} \right\rceil$ (always ceiling/round up).
3. Line Efficiency: $E = \frac{T_{wc}}{N \cdot T_c} \times 100$.
4. Balance Delay (Lost Time): $d = 100$.

Comprehensive Summary: Exam Formulas Quick-Reference

Metric	Governing Formula	Key Physical Units	Exam Gotcha / Trap
Break-Even Volume	$Q_{BEP} = \frac{FC}{P-v}$	units / year	Forgetting to subtract variable cost v from selling price P .
Target Selling Price	$P_{\min} = v + \frac{FC}{Q}$	\$ / unit	Forgetting that price must cover both variable cost and amortized fixed cost.

Metric	Governing Formula	Key Physical Units	Exam Gotcha / Trap
Crossover Quantity	$Q_{1,2} = \frac{FC_2 - FC_1}{v_1 - v_2}$	units / year	Subtracting in the wrong direction (FC high — low, v high — low).
Weighted Center of Gravity	$x^* = \frac{\sum W_i x_i}{\sum W_i}, y^* = \frac{\sum W_i y_i}{\sum W_i}$	km, miles, or grid units	Forgetting to divide by total weight $\sum W_i$.
Manhattan Distance	$d_R = x_1 - x_2 + y_1 - y_2 $	km or miles	Squaring terms like Euclidean distance instead of taking absolute sums.
Cycle Time	$T_c = T_m + T_h + T_t$	min / piece	Confusing cycle time with total shift operating time.
Production Rate	$R_p = \frac{60}{T_c}$	pieces / hour	Leaving T_c in seconds without converting to minutes.
Machine Availability	$A = \frac{T_{operating}}{T_{planned}}$	dimensionless (%)	Including scheduled lunch/PM in the unplanned downtime denominator.
Min Workstations	$N_{\min} = \left\lceil \frac{T_{wc}}{T_c} \right\rceil$	integer stations	Rounding down instead of rounding UP to the next integer.
Line Efficiency	$E = \frac{T_{wc}}{N \cdot T_c}$	dimensionless (%)	Using $N_{\min}$ instead of the actual assigned stations N .